

Bowen QIU

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EDUCATION

Zhejiang University (ZJU), Hangzhou, China

Sept. 2021 – Jun. 2025

- **B.E. in Biomedical Engineering**, Minor in **Finance**.
- GPA: **3.99/4.00**; Weighted Average Score: **92.58/100**; Ranking: **1/133**.

Oxford Prospects Interdisciplinary Academic Programme, Oxford, UK

Aug. 2024

- Completed a two-week interdisciplinary academic programme in Oxford, UK.
- Received an Excellent (A-) grade in the Interdisciplinary Academic module.

The University of Hong Kong (HKU), Hong Kong, China

Sept. 2025 – Jun. 2029

- **PhD in the Department of Electrical and Computer Engineering, Faculty of Engineering.**

SCHOLARSHIPS & AWARDS

The CHU KO CHEN scholarship in ZJU

Oct. 2024

- The top scholarship by ZJU. 12 out of 6,000 undergraduates are awarded with the scholarship year.

The National Scholarship

2021-2022, 2022-2023 & 2023-2024

- Established by the Ministry of Education. Approximately 2% students in ZJU are awarded per year.

The TOP-10 Outstanding Students in ZJU

Dec. 2024

- The highest awards in ZJU. Only 10 out of 60,000 students are awarded per year.

Hong Kong PhD Fellowship Scheme (HKPFS) and HKU Presidential PhD Scholars (HKUPS)

Sept. 2025

- HKPFS is established by Hong Kong government, granting no more than 400 students annually.

RESEARCH ASSISTANT

Harvard Medical School / Boston Children's Hospital, Boston, US

Jul. 2024 - Nov. 2024

- Mentors: Professor Kiho Im, Dr. Sungmin You, and Dr. Andrea Gondova.
- Worked on regional analysis of error patterns in automatic cortical plate segmentation in fetal brain MRI. Developed a method to identify brain regions with the largest errors and compared the performance of pretrained UNet and fuzzy-UNet models.
- Got Proficiency in Linux system and gain experience of working with large scientific research group.
- Presented this work as a digital poster at OHBM 2025.

PAPERS, PATENT & COPYRIGHTS

1. **Bowen Qiu**, Rui Qian, Baorong Gu, Zichao Li, Zhifan Chen, Xinyi Xu, Huaijin Gao, Yiwei Chen, Ruoke Zhao, Ruike Chen, Yuqi Zhang, Lingxuan Zhang, Zhiyong Zhao, Mingyang Li*, and Dan Wu*. Neural correlates differ between crystallized and fluid intelligence in adolescents. *Translational Psychiatry* 15, 246 (2025).

<https://doi.org/10.1038/s41398-025-03467-4>

2. Huaijin Gao, Yihan Ma, Rui Qian, Tao Jin, Baorong Gu, **Bowen Qiu**, Shaoyong Ye, Manli Huang, Dan Wu, and Zhiyong Zhao. Distinct and Shared Molecular Mechanisms Underlie Morphological-Functional Overcoupling and Undercoupling in Major Depressive Disorder. *Biological Psychiatry* (2026).

<https://doi.org/10.1016/j.biopsych.2026.06.016>

3. **Bowen Qiu**, Rui Qian, Baorong Gu, Zichao Li, Zhifan Chen, Xinyi Xu, Yiwei Chen, Ruoke Zhao, Ruike Chen, Yuqi Zhang, Zhiyong Zhao, Mingyang Li*, and Dan Wu*. Neural correlates differ between crystallized and fluid intelligence in adolescents: A Multi-modal MRI Investigation, *ISMRM 2025*, Digital Poster.

4. **Bowen Qiu**, Andrea Gondova, Sungmin You, Milton O. Candela-Leal, Hanuman Verma, Seungyoon Jeong, P. Ellen Grant and Kiho Im*. Regional Analysis of Error Patterns from Automatic Cortical Plate Segmentation in Fetal Brain MRI, *OHBM 2025*, Digital Poster.

5. Rui Qian, Huaijin Gao, **Bowen Qiu**, Zichao Li, Baorong Gu, Tongmei Ye, Dan Wu*, and Zhiyong Zhao*. Multi-Scale Characterization of Distance-dependent Functional Dysconnections in Major Depressive Disorder Across Two Asian Cohorts: A Genomic, Neurochemical, and Cellular Perspective, *MedRxiv*, <https://medrxiv.org/cgi/content/short/2024.11.20.24317665v1>.

6. Lingxuan Zhang, **Bowen Qiu**, Zhifan Chen, Xinyi Xu, Ruoke Zhao, Yiwei Chen, Chenglin Ning, Ruike Chen, Mingyang Li, Dan Wang, Junfen Fu, and Dan Wu. Longitudinal Central Adiposity Accumulation is Associated with Cortical Alteration and Impaired Cognitive Function in Adolescents. *eLife* 15:RP111663 (2026), Reviewed Preprint v1. <https://doi.org/10.7554/eLife.111663.1>
7. Rui Qian, Huaijin Gao, **Bowen Qiu**, Baorong Gu, Zichao Li, Dan Wu*, and Zhiyong Zhao*. Distance-Dependent Functional Dysconnectivity and Neurotransmitter Correlates in Major Depressive Disorder, *ISMRM 2025*, Oral.
8. Baorong Gu, Rui Qian, Zilong Ye, **Bowen Qiu**, Zichao Li, Mingyang Li, Ruoke Zhao, Yuqi Zhang, Yutian Wang, Peiying Liu, Hanzhang Lu, Dan Wu*, and Zixuan Lin*. Cerebrovascular Reactivity Development in Infants during the First Two Years of Life Using Resting-State BOLD Functional MRI, *ISMRM 2025*. Digital Poster.
9. **Bowen Qiu**, Vick Lau, Xuehong Lin, Junhao Zhang, Xiang Li, Gilberto K. K. Leung, Yujiao Zhao, Alex T. L. Leong, and Ed X. Wu. Human Brain Age Prediction using 0.055T Ultra-low-field MRI. *ISMRM 2026*, Digital Poster.
10. Baorong Gu, Rui Qian, Zilong Ye, **Bowen Qiu**, Zixuan Lin* and Dan Wu*. Cerebrovascular Reactivity Mapping Using Resting-state Functional MRI in Infants Aged 0 to 6 Months”, *OHBM 2024*, Digital Poster.
11. Xinlin Yu, **Bowen Qiu**, Xuan Xie, Jinsong Wu, Yihan Chen, Baorong Gu, Lingxuan Zhang, Ning Deng*. A smart toilet with seat for health monitoring, Chinese patent application No. 202410326127.6.
12. Ning Deng*, **Bowen Qiu**, Lingxuan Zhang, Jinsong Wu, Xuan Xie. Yihan Chen. Xinlin Yu. Intestinal Health Assistant - a WeChat Mini Program, Registered Software Copyright: 2023SR1496402.
13. Rui Qian, **Bowen Qiu**, Baorong Gu, Zichao Li. Leheng Qiu, Dan Wu. Toolbox for mapping sub-atlases of the brain [Atlas2vec], Registered Software Copyright: 2024SR1598299.

RESEARCH EXPERIENCE

Study of Obesity-brain-cognitive Correlation in Adolescents Based on MRI Brain Imaging | Leader of National University Student Innovation & Entrepreneurship Development Program **Mar. 2023 - Jun. 2024**

- Supervisors: Professor Dan Wu and Dr. Mingyang Li. Research fund: 12,000 CNY.
- Supervised the research process and took charge of organizing and reporting.
- Analyzed the data using tools such as MATLAB, RStudio, and Python.
- Second author of a 2026 *eLife* Reviewed Preprint on central adiposity, cortical changes, and cognition.

Neural and Genetic Basis Exploration of Adolescent Intelligence Development Based on Adolescent Brain Cognitive Development (ABCD) study | Individual Project **Nov. 2023 - Jul. 2024**

- Supervisors: Professor Dan Wu and Dr. Mingyang Li. Research fund: 20,000 CNY.
- The project is awarded as project of ZJU undergraduate rising scientific rising star. (30 projects annually).
- Utilized data from ABCD and Allen Brain datasets to investigate the distinct aspects of fluid/crystallized intelligence, employing tools of spatial correlation and gene enrichment.
 - Got proficiency in toolboxes such as MATLAB, RStudio, and Python through data processing and analysis tasks.
 - First-author article published in *Translational Psychiatry* (2025); findings presented at *ISMRM 2025*.

Intestinal Health Assistant Software Development | Leader of a 7-member team **Jul. 2023 - Mar. 2024**

- Developed a procedure enabling patients to record and upload stool properties, such as weight and color, to realize intestinal disease diagnosis. The functions are combined in the We-chat software.
 - Took responsibility for realizing the uploading, downloading and local storage of patient data.
 - Oversaw the development process, ensuring quality control for each task and the overall group performance.
- Obtained a software copyright registration and contributed to a patent application.

Brain Tumor Image Classification Based on ResNet | Project for Medical AI course **April 2024**

- Adapted ResNet50 for Brain Tumor Classification through Transfer Learning.
- Added various pre-processing methods to improve the deep learning model for Tumor Classification.

OTHER ACADEMIC ACTIVITIES

The Chinese University of Hong Kong, Hong Kong **Jul. 2024**

- Participated in CUHK Faculty of Engineering PhD Summer Workshop.

The Hong Kong Polytechnic University, Hong Kong **Jan. 2024**

- Participated in the Exchange Program visiting to Professor Anqi Qiu's Research Team in PolyU.

EXTRACURRICULAR ACTIVITIES

Leader of Social Practice Group "Best Converges Brain Bank Center"

Summer 2022 & 2023

- Initiated the group and raised funding for social practice activities, under the guidance of Prof Aimin Bao.
- Planned brain bank presentations and led the group to visit Peking Union Medical College Brain Bank, Shanghai Fudan University Medical School Brain Bank, etc.
- Wrote a report on the status of brain bank development in China and consider methods to promote brain bank development from the perspective of students.
- Awarded the ZJU Advanced Individual in Social Practice for 2023 Summer Vacation, and the ZJU Outstanding Team in Social Practice for 2022 and 2023 Summer Vacation.

Student Chairman of Zhejiang University Wind Orchestra

Mar. 2023 - May 2024

- Formulated policies related to the orchestra. Coordinate the recruitment of new members, daily attendance, and financial reimbursement of the orchestra. Promoted communications with other orchestras at ZJU.
- Planned and organized three exclusive concerts (March 2023, Oct. 2023 and May 2024), led the team to present perform on 2024 New Year's Eve.
- Led the team to win the first prize in both the Instrumental Ensemble Competition at the University Students Art Festival in Zhejiang Province and the Instrumental Competition in the Yangtze River Delta Region, China.

Teaching Assistant for Lower Grade Students

Sept. 2022 - Jun. 2023

- Organized academic tutoring activities and research guidance for lower-grade students, gave multiple lectures on how to lay a solid foundation in academic subjects at university.
- Worked as a teaching assistant for a Calculus class, delivering lessons, grading homework, and providing support by answering students' questions outside of class.
- Awarded the "Outstanding Senior" and "Star of Peer Tutoring" title by Zhejiang University.

Participant of "Riding Waves, Chasing Dreams: Snow Dragon Ship" Social Practice

Aug. 2023

- Conducted scientific research and experienced life at sea aboard the Snow Dragon Ship.
- Collected organism from the deep ocean and observed them under a microscope.
- Handled supply management tasks as the leader of the logistics working group during the social practice.
- Awarded the ZJU Outstanding Team in Social Practice for 2023 Summer Vacation.

Main Organizer of Alumni Visiting Social Practice

Aug. 2023

- Visited alumni in Xi'an and Chengdu City, furthering connection between university and alumni association.
- Confirmed destination and arranged rundown for the social practice as the main organizer of the social practice.
- Awarded the Outstanding Team in ZJU Global Alumni Visiting for 2023 Summer Vacation.

ACADEMIC COMPETITIONS

- The 3rd Prize, the National College Biomedical Engineering Innovation Design Competition (July 2024)
- The 1st Prize, National Undergraduate Mathematics Competition (Non-Mathematics Major) (Jan. 2023)
- The 1st Prize, Zhejiang Provincial College Student Physics Innovation (Theoretical) Competition (Jan. 2023)
- Bronze Medal in Zhejiang University "Internet+" Innovation and Entrepreneurship Competition (Jun. 2023)
- The 1st Prize, the Creativity Collection Competition held by CHUANHUA Co., Ltd. (May 2023)
- The 3rd Prize, College Students' Science and Technology Competition for Energy Conservation and Emission Reduction held by Zhejiang University (May 2023)

OTHER INFORMATION

- Programming Skills: Python, MATLAB, RStudio, JavaScript and C.
- Hobbies: trumpet, video clipping, swimming and table tennis.